SCSI1113: COMPUTATIONAL MATHEMATICS
2021-2022/Semester 1
Assignment 3
Deadline : 18 Disember 2021 2pm

1)

a) Let $W = \{(x, y, z) \in \Re^3 | 3x = 2y\}$. Prove that W is subspace of $\Re^3$.

b) Verify W is subspace of $V \in M_{3x2}$. W is the set of all 3x2 matrices of the form of,

$$\begin{bmatrix} a & b \\ a+b & 0 \\ 0 & c \end{bmatrix}$$

2) Let $S = \{(4,7,3), (-1,2,6), (2,-3,5)\}$, Is S spanning in $\Re^3$?

3) Determine the basis and dimension of the solution space of the homogenous system,

$$2x_1 + 2x_2 + 4x_3 - 2x_4 = 0$$
$$x_1 + 2x_2 + x_3 + 2x_4 = 0$$
$$-x_1 + x_2 + 4x_3 - 2x_4 = 0$$

4) For the following matrix, calculate the rank, determine the dimension of the column space and provide a basis for the column space :

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 4 \\ 3 & -1 & 2 \end{bmatrix}$$

5) Given ordered basis $B$and $B'$and given $[v]_B$ find

$$B = \left\{\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right\} B' = \left\{\begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\right\}$$

$$[x]_B = \begin{bmatrix} 6 \\ -1 \\ -1 \end{bmatrix}$$

a) Transition matrix B to B'

b) $[x]_{B'}$